



CODEIGNITER CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & CONFIG

Installation

```
composer create-project codeigniter4/apstart...  
php spark serve  
.env: CI_ENVIRONMENT = development  
.env: database.default.hostname = localhost  
app/Config/Routes.php for routing configurati...
```

04. MODELS

Workflow

```
class UserModel extends Model {  
    protected $table = 'users';  
    protected $allowedFields = ['name', 'email'];  
    protected $useTimestamps = true;  
    protected $validationRules = ['email' => 'req...'  
]  
    $user = $model->find(1);  
    $model->save(['name'=>'Alice', 'email'=>'a@b.c...'  
    $model->where('active', 1)->findAll();
```

02. ROUTING

Architecture

```
$routes->get('/users', 'UserController::index...'  
$routes->post('/users', 'UserController::crea...'  
$routes->resource('users'); // RESTful resou...  
$routes->get('/users/{:num}', 'UserController...'  
$routes->group('api', function($routes) { .....
```

05. VIEWS

CRUD API

```
// In controller: return view('users/index', ...  
<?= esc($user->name) ?>  
<?php foreach ($users as $user): ?>  
<li><?= esc($user->name) ?></li>  
<?php endforeach ?>  
<?= view('partials/header') ?>  
<?= form_open('users/create') ?>  
<?= form_input('name', '', ['class'=>'form-co...
```

03. CONTROLLERS

YAML / Config

```
class UserController extends BaseController {  
    public function index(): string {  
        $model = new UserModel();  
        $data = ['users' => $model->findAll()];  
        return view('users/index', $data);  
    }  
    public function show(int $id): ResponseInterf...  
    return $this->response->setJSON(model(UserMod...  
    }  
}
```

06. QUERY BUILDER

Integration

```
$builder = $db->table('users');  
$query = $builder->select('name,email')  
->where('active', 1)  
->orderBy('name')  
->get(10, 0); // limit 10, offset 0  
$results = $query->getResultArray();  
$builder->insert(['name'=>'Alice', 'email'=>'a...'  
$db->transStart(); ... $db->transComplete();
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. FORM VALIDATION

Hardening

```
$rules = [  
'name' => 'required|min_length[2]|max_length[...  
'email' => 'required|valid_email|is_unique[us...  
];  
if (!$this->validate($rules)) {  
    return redirect()->back()->withInput()  
->with('errors', $this->validator->getErrors(...  
}
```

10. COMMON HELPERS

Unit Tests

```
url_title($str) // 'My Article' 'my-articl...  
word_wrap($str, 76)  
excerpt($str, $phrase, $radius)  
character_limiter($str, 100)  
number_to_currency($n, 'USD')  
php spark make:controller/model/migration/ent...
```

08. LIBRARIES & HELPERS

Observability

```
$email = \Config\Services::email();  
$email->setTo('a@b.com'); $email->send();  
// Load helper functions:  
helper(['url', 'form', 'text']);  
base_url('assets/css/main.css')  
site_url('users') // includes index.php if n...
```

11. COMMON GOTCHAS

Commands

\$allowedFields prevents mass assignment vulnerabilities
esc() everything from user input prevent XSS
CSRF protection enabled by default for POST requests
Remove index.php from URL: configure RewriteBase in .htaccess
Config files in app/Config don't edit vendor files

09. SESSIONS

Optimization

```
$session = session();  
$session->set('user_id', $user->id);  
$session->get('user_id');  
$session->remove('user_id');  
$session->destroy();  
// Flash data (one-time):  
$session->setFlashdata('success', 'Saved!');  
$session->getFlashdata('success');
```